Study 3: Group-level perceptions and metaperceptions

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Configuration

```
data_path <- "/Users/yaldasmac/Desktop/Metaperception/Study\ 3/study3_data.xlsx"

id_col       <- "ID"
ideology_col <- "ideology"
group_codes     <- c(Liberals = 1, Conservatives = 2)
condition_codes <- c(Outgroup = 1, Metaperception = 2, Ingroup = 3)

# MFQ-2 families
binding_subscales <- list(
  loyalty         = c("loyalty_1", "loyalty_2"),
  authority       = c("authority_1", "authority_2"),
  purity          = c("purity_1", "purity_2"),
  proportionality = c("proportionality_1", "proportionality_2")
)
individualizing_subscales <- list(
  care     = c("care_1", "care_2"),
  equality = c("equality_1", "equality_2")
)
```

```r
library(readxl)
library(dplyr)
library(tidyr)
library(ggplot2)
library(car)
library(emmeans)
library(effectsize)
library(knitr)
has_heplots <- requireNamespace("heplots", quietly = TRUE)
```

```r
fmt_p <- function(p) ifelse(p < .001, "< .001", sub("^0", "", sprintf("%.3f", p)))
fmt_n <- function(x, d = 2) sub("^(-?)0\\.", "\\1.", sprintf(paste0("%.", d, "f"), x))

# Contrast weights keyed to an emmGrid's own row order
weights_for <- function(emm, pos, neg) {
  g <- as.data.frame(emm)
  lab <- paste(g$group, g$condition, sep = ".")
  w <- setNames(rep(0, length(lab)), lab)
  stopifnot(pos %in% lab, neg %in% lab)
  w[pos] <- 1
  w[neg] <- -1
  w
}
```

1. Load data and build composites

Composites follow `morality`: each subscale is the mean of its two items, and each family is the mean of its subscales. With complete data this equals the item mean.

```r
raw <- read_excel(data_path)

needed <- c(unlist(binding_subscales), unlist(individualizing_subscales),
            ideology_col, "Group", "Condition")
miss <- setdiff(needed, names(raw))
if (length(miss)) stop("Missing columns: ", paste(miss, collapse = ", "))

sub_mean <- function(df, items) rowMeans(sapply(df[items], as.numeric), na.rm = TRUE)

data <- raw %>%
  mutate(
    group = factor(names(group_codes)[match(Group, group_codes)],
                   levels = names(group_codes)),
    condition = factor(names(condition_codes)[match(Condition, condition_codes)],
                       levels = names(condition_codes)),
    ideology = as.numeric(.data[[ideology_col]])
  )

data$binding <- rowMeans(sapply(binding_subscales, sub_mean, df = data), na.rm = TRUE)
data$individualizing <- rowMeans(sapply(individualizing_subscales, sub_mean, df = data),
                                 na.rm = TRUE)

data <- data %>% filter(!is.na(group), !is.na(condition))
data$ideology_c <- data$ideology - mean(data$ideology, na.rm = TRUE)

if (id_col %in% names(data)) {
  cat("Duplicate IDs:", sum(duplicated(data[[id_col]])), "\n")
}
```

```
Duplicate IDs: 0
```

```r
kable(with(data, table(group, condition)), caption = "Cell sizes")
```

Table 1: Cell sizes

	Outgroup	Metaperception	Ingroup
Liberals	100	95	99
Conservatives	94	86	92

2. Descriptives (raw, unadjusted)

```
desc <- data %>%
  group_by(group, condition) %>%
  summarise(n = n(),
            M_ind = mean(individualizing, na.rm = TRUE),
            SD_ind = sd(individualizing, na.rm = TRUE),
            M_bind = mean(binding, na.rm = TRUE),
            SD_bind = sd(binding, na.rm = TRUE),
            M_ideology = mean(ideology, na.rm = TRUE),
            .groups = "drop")
kable(desc, digits = 3)
```

group	condition	n	M_ind	SD_ind	M_bind	SD_bind	M_ideology
Liberals	Outgroup	100	2.212	0.975	3.288	0.961	2.130
Liberals	Metaperception	95	3.042	1.012	2.004	0.674	2.158
Liberals	Ingroup	99	3.790	0.723	2.658	1.069	2.384
Conservatives	Outgroup	94	3.309	0.778	2.739	1.089	5.660
Conservatives	Metaperception	86	3.058	0.754	3.565	0.782	5.663
Conservatives	Ingroup	92	3.033	0.785	3.409	0.791	5.728

The `M_ideology` column shows how far apart the groups sit on the covariate. An ideology-adjusted mean is evaluated at the grand mean (about the scale midpoint), a value few participants in either group actually hold.

3. Assumption checks

```
# Homogeneity of regression slopes: does ideology relate to ratings
# differently across cells?
slopes_ind <- lm(individualizing ~ ideology_c * group * condition, data = data)
slopes_bin <- lm(binding ~ ideology_c * group * condition, data = data)
kable(Anova(slopes_ind, type = 2), digits = 3,
      caption = "Individualizing: covariate x factor interactions (Type II)")
```

Table 2: Individualizing: covariate x factor interactions (Type II)

	Sum Sq	Df	F value	Pr(>F)
ideology_c	0.815	1	1.159	0.282
group	0.136	1	0.193	0.660
condition	43.470	2	30.918	0.000
ideology_c:group	0.723	1	1.029	0.311
ideology_c:condition	1.916	2	1.363	0.257
group:condition	22.951	2	16.323	0.000
ideology_c:group:condition	9.058	2	6.442	0.002
Residuals	389.458	554	NA	NA

```
kable(Anova(slopes_bin, type = 2), digits = 3,
      caption = "Binding: covariate x factor interactions (Type II)")
```

Table 3: Binding: covariate x factor interactions (Type II)

	Sum Sq	Df	F value	Pr(>F)
ideology_c	3.894	1	4.859	0.028
group	10.885	1	13.585	0.000
condition	8.600	2	5.367	0.005
ideology_c:group	6.414	1	8.005	0.005
ideology_c:condition	9.837	2	6.138	0.002
group:condition	22.133	2	13.812	0.000
ideology_c:group:condition	0.940	2	0.587	0.557
Residuals	443.895	554	NA	NA

```
if (has_heplots) {
  print(heplots::boxM(cbind(individualizing, binding) ~ group * condition, data = data))
} else {
  cat("Install 'heplots' to run Box's M.\n")
}
```

```
 Box's M-test for Homogeneity of Covariance Matrices

data:  data
Chi-Sq (approx.) = 71.7776, df = 15, p-value = 2.151e-09
```

Significant `ideology_c:condition` terms mean the single-slope ANCOVA assumption is violated. Section 8 checks whether that matters for the planned contrasts.

4. MANCOVA

Sum-to-zero contrasts and a centered covariate make every Type III test interpretable. The first table uses `car::Manova` (Type III). The second reproduces the sequential `manova()` call from `morality`, whose interaction test is what the manuscript reports.

```
options(contrasts = c("contr.sum", "contr.poly"))

mlm_fit <- lm(cbind(individualizing, binding) ~ group * condition + ideology_c,
              data = data)
mancova_III <- Manova(mlm_fit, type = 3)
print(mancova_III, test = "Pillai")
```

```
Type III MANOVA Tests: Pillai test statistic
               Df test stat approx F num Df den Df    Pr(>F)
(Intercept)     1   0.95183   5513.6      2    558 < 2.2e-16 ***
group           1   0.01703      4.8      2    558   0.00829 **
condition       2   0.10848     16.0      4   1118 9.055e-13 ***
ideology_c      1   0.00907      2.6      2    558   0.07876 .
group:condition 2   0.37097     63.6      4   1118 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
manova_seq <- manova(cbind(individualizing, binding) ~ group * condition + ideology,
                     data = data)
summary(manova_seq, test = "Pillai")
```

```
                Df  Pillai approx F num Df den Df    Pr(>F)
group            1 0.08948   27.420      2    558 4.378e-12 ***
condition        2 0.12367   18.422      4   1118 1.134e-14 ***
ideology         1 0.00923    2.599      2    558   0.07527 .
group:condition  2 0.37097   63.649      4   1118 < 2.2e-16 ***
Residuals      559
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

5. Univariate ANCOVAs (corrected)

```
fit_ind <- lm(individualizing ~ group * condition + ideology_c, data = data)
fit_bin <- lm(binding ~ group * condition + ideology_c, data = data)

ancova_table <- function(fit, label) {
  a <- Anova(fit, type = 3)
  a <- a[rownames(a) != "(Intercept)", ]
  eta <- eta_squared(Anova(fit, type = 3), partial = TRUE, ci = .95)
  res_df <- a["Residuals", "Df"]
  a <- a[rownames(a) != "Residuals", ]
  data.frame(DV = label, effect = rownames(a),
             F = a$`F value`, df1 = a$Df, df2 = res_df,
             p = fmt_p(a$`Pr(>F)`),
             eta2_p = eta$Eta2_partial[match(rownames(a), eta$Parameter)],
             row.names = NULL)
}

ancova_corrected <- rbind(ancova_table(fit_ind, "Individualizing"),
                          ancova_table(fit_bin, "Binding"))
kable(ancova_corrected, digits = 3,
      caption = "Type III ANCOVA, sum contrasts, centered ideology")
```

Table 4: Type III ANCOVA, sum contrasts, centered ideology

DV	effect	F	df1	df2	p	eta2_p
Individualizing	group	0.038	1	559	.846	0.000
Individualizing	condition	28.003	2	559	< .001	0.091
Individualizing	ideology_c	1.135	1	559	.287	0.002
Individualizing	group:condition	57.728	2	559	< .001	0.171
Binding	group	9.483	1	559	.002	0.017
Binding	condition	4.105	2	559	.017	0.014
Binding	ideology_c	4.717	1	559	.030	0.008
Binding	group:condition	64.712	2	559	< .001	0.188

For comparison, the original specification (treatment contrasts, uncentered ideology). Only the `group:condition` and `ideology` rows should match the corrected table; the `group` and `condition` rows were simple effects, not main effects.

```
options(contrasts = c("contr.treatment", "contr.poly"))
fit_ind_old <- lm(individualizing ~ group * condition + ideology, data = data)
fit_bin_old <- lm(binding ~ group * condition + ideology, data = data)
old_tab <- rbind(
  data.frame(DV = "Individualizing", effect = rownames(Anova(fit_ind_old, type = 3)),
             F = Anova(fit_ind_old, type = 3)$`F value`),
  data.frame(DV = "Binding", effect = rownames(Anova(fit_bin_old, type = 3)),
             F = Anova(fit_bin_old, type = 3)$`F value`)
)
kable(subset(old_tab, !effect %in% c("(Intercept)", "Residuals")), digits = 2,
      row.names = FALSE, caption = "Original (treatment-contrast) Type III F values")
```

Table 5: Original (treatment-contrast) Type III F values

DV	effect	F
Individualizing	group	42.92
Individualizing	condition	85.35
Individualizing	ideology	1.14
Individualizing	group:condition	57.73
Binding	group	21.73
Binding	condition	48.75
Binding	ideology	4.72
Binding	group:condition	64.71

```
options(contrasts = c("contr.sum", "contr.poly"))
```

within-group test that each group's error is larger on one foundation.

Delta_ind - Delta_bind = (meta_I - meta_B) - (actual_I - actual_B)

```
# #6: within-group test that each group's error is larger on one foundation.
# Delta_ind - Delta_bind = (meta_I - meta_B) - (actual_I - actual_B)
data$D <- data$individualizing - data$binding
for (f in list(D ~ group * condition + ideology_c, D ~ group * condition)) {
  emm_D <- emmeans(lm(f, data = data), ~ group * condition)
  print(summary(contrast(emm_D, method = list(
    "Cons: d_ind - d_bind" = weights_for(emm_D, "Conservatives.Metaperception", "Liberals.Outgroup"),
    "Libs: d_ind - d_bind" = weights_for(emm_D, "Liberals.Metaperception", "Conservatives.Outgroup")),
    adjust = "none"), infer = TRUE))
}
```

```
 contrast             estimate    SE  df lower.CL upper.CL t.ratio p.value
 Cons: d_ind - d_bind    0.684 0.202 559   0.2865    1.081   3.381  0.0008
 Libs: d_ind - d_bind    0.354 0.200 559  -0.0391    0.747   1.769  0.0775

Confidence level used: 0.95
 contrast             estimate    SE  df lower.CL upper.CL t.ratio p.value
 Cons: d_ind - d_bind    0.568 0.163 560    0.248    0.888   3.487  0.0005
 Libs: d_ind - d_bind    0.469 0.161 560    0.153    0.785   2.912  0.0037

Confidence level used: 0.95
```

```
# Both estimates should be positive. With the preregistered covariate the
# point estimates are 0.68 (conservatives) and 0.35 (liberals).
```

6. Estimated marginal means

Adjusted to the grand-mean ideology.

```
emm_ind <- emmeans(fit_ind, ~ group * condition)
emm_bin <- emmeans(fit_bin, ~ group * condition)
kable(as.data.frame(emm_ind), digits = 3, caption = "Individualizing EMMs")
```

Table 6: Individualizing EMMs

group	condition	emmean	SE	df	lower.CL	upper.CL
Liberals	Outgroup	2.261	0.096	559	2.072	2.450
Conservatives	Outgroup	3.259	0.099	559	3.065	3.453
Liberals	Metaperception	3.090	0.098	559	2.898	3.282
Conservatives	Metaperception	3.009	0.102	559	2.808	3.210
Liberals	Ingroup	3.832	0.094	559	3.648	4.016
Conservatives	Ingroup	2.982	0.100	559	2.784	3.179

```
kable(as.data.frame(emm_bin), digits = 3, caption = "Binding EMMs")
```

Table 7: Binding EMMs

group	condition	emmean	SE	df	lower.CL	upper.CL
Liberals	Outgroup	3.394	0.103	559	3.191	3.596
Conservatives	Outgroup	2.632	0.106	559	2.424	2.840
Liberals	Metaperception	2.109	0.105	559	1.902	2.315
Conservatives	Metaperception	3.458	0.110	559	3.242	3.674
Liberals	Ingroup	2.749	0.100	559	2.551	2.946
Conservatives	Ingroup	3.298	0.108	559	3.086	3.509

7. Planned contrasts (preregistered)

Unadjusted p values, as in `morality`. `d_gap` is the standardized difference between the two contrasts in a family (the manuscript's between-group gap in accuracy).

```
run_contrasts <- function(emm, fit, pairs, family) {
  wl <- lapply(pairs, function(pr) weights_for(emm, pr[1], pr[2]))
  ct <- contrast(emm, method = wl, adjust = "none")
  s  <- summary(ct, infer = c(TRUE, TRUE))
  sig <- sigma(fit)
  out <- data.frame(family = family, contrast = s$contrast,
                    M_diff = s$estimate, SE = s$SE,
                    CI_low = s$lower.CL, CI_high = s$upper.CL,
                    t = s$t.ratio, df = s$df, p = fmt_p(s$p.value),
                    d = s$estimate / sig)
  gap <- summary(eff_size(ct, sigma = sig, edf = df.residual(fit)))
  list(table = out, gap = data.frame(family = family,
                                     d_gap = gap$effect.size,
                                     CI_low = gap$lower.CL, CI_high = gap$upper.CL),
       ct = ct)
}

h1_pairs <- list(
  "Liberal ingroup vs Conservative outgroup"   = c("Liberals.Ingroup", "Conservatives.Outgroup"),
  "Conservative ingroup vs Liberal outgroup"   = c("Conservatives.Ingroup", "Liberals.Outgroup"))
h2_pairs <- list(
  "Liberal outgroup vs Conservative metaperception" = c("Liberals.Outgroup", "Conservatives.Metaperception"),
  "Conservative outgroup vs Liberal metaperception" = c("Conservatives.Outgroup", "Liberals.Metaperception"))

h1_ind <- run_contrasts(emm_ind, fit_ind, h1_pairs, "Individualizing")
h1_bin <- run_contrasts(emm_bin, fit_bin, h1_pairs, "Binding")
h2_ind <- run_contrasts(emm_ind, fit_ind, h2_pairs, "Individualizing")
h2_bin <- run_contrasts(emm_bin, fit_bin, h2_pairs, "Binding")

kable(rbind(h1_ind$table, h1_bin$table), digits = 3,
      caption = "H1: ingroup vs outgroup perception (opposing stance)")
```

Table 8: H1: ingroup vs outgroup perception (opposing stance)

family	contrast	M_diff	SE	CI_low	CI_high	t	df	p	d
Individualizing	Liberal ingroup vs Conservative outgroup	0.573	0.149	0.280	0.865	3.849	559	< .001	0.676
Individualizing	Conservative ingroup vs Liberal outgroup	0.721	0.154	0.418	1.023	4.678	559	< .001	0.850
Binding	Liberal ingroup vs Conservative outgroup	0.117	0.160	-0.197	0.430	0.731	559	.465	0.128
Binding	Conservative ingroup vs Liberal outgroup	-0.096	0.165	-0.421	0.228	-0.583	559	.560	-0.106

```
kable(rbind(h1_ind$gap, h1_bin$gap), digits = 3, caption = "H1: between-group gap (d)")
```

Table 9: H1: between-group gap (d)

family	d_gap	CI_low	CI_high
Individualizing	-0.175	-0.751	0.402
Binding	0.234	-0.342	0.810

```
kable(rbind(h2_ind$table, h2_bin$table), digits = 3,
      caption = "H2: outgroup perception vs metaperception (the accuracy tests)")
```

Table 10: H2: outgroup perception vs metaperception (the accuracy tests)

family	contrast	M_diff	SE	CI_low	CI_high	t	df	p	d
Individualizing	Liberal outgroup vs Conservative metaperception	-0.748	0.155	-1.052	-0.444	-4.833	559	< .001	-0.883
Individualizing	Conservative outgroup vs Liberal metaperception	0.169	0.153	-0.131	0.470	1.106	559	.269	0.200
Binding	Liberal outgroup vs Conservative metaperception	-0.064	0.166	-0.390	0.262	-0.387	559	.699	-0.071
Binding	Conservative outgroup vs Liberal metaperception	0.524	0.164	0.201	0.846	3.187	559	.002	0.576

```
kable(rbind(h2_ind$gap, h2_bin$gap), digits = 3, caption = "H2: between-group gap (d)")
```

Table 11: H2: between-group gap (d)

family	d_gap	CI_low	CI_high
Individualizing	-1.083	-1.671	-0.494
Binding	-0.647	-1.235	-0.059

Table 1 rows (Δ = metaperception − actual)

The emmeans contrast is *actual* − *metaperception*, so Δ flips its sign.

```
t1 <- rbind(h2_ind$table, h2_bin$table)
t1$judge <- ifelse(grepl("Conservative metaperception", t1$contrast),
                   "Conservative", "Liberal")
t1$Delta <- -t1$M_diff
t1$reading <- ifelse(t1$p != "< .001" & as.numeric(sub("< ", "", t1$p)) >= .05,
                     "Accurate", ifelse(t1$Delta > 0, "Overestimate", "Underestimate"))
kable(t1[, c("judge", "family", "Delta", "SE", "t", "df", "p", "reading")],
      digits = 2, row.names = FALSE)
```

judge	family	Delta	SE	t	df	p	reading
Conservative	Individualizing	0.75	0.15	-4.83	559	< .001	Overestimate

judge	family	Delta	SE	t	df	p	reading
Liberal	Individualizing	-0.17	0.15	1.11	559	.269	Accurate
Conservative	Binding	0.06	0.17	-0.39	559	.699	Accurate
Liberal	Binding	-0.52	0.16	3.19	559	.002	Underestimate

8. Robustness of the planned contrasts

Three specifications of the H2 accuracy contrasts:

- **A. Preregistered:** single ideology slope, means at grand-mean ideology (section 7).
- **B. No covariate:** raw cell means.
- **C. Separate slopes:** ideology allowed to relate differently to ratings in each condition, with each group evaluated at its own mean ideology.

```
h2_under <- function(fit, emm, family, spec) {
  r <- run_contrasts(emm, fit, h2_pairs, family)$table
  r$spec <- spec
  r$Delta <- -r$M_diff
  r[, c("spec", "family", "contrast", "Delta", "SE", "t", "p")]
}

# B: no covariate
fit_ind_B <- lm(individualizing ~ group * condition, data = data)
fit_bin_B <- lm(binding ~ group * condition, data = data)

# C: separate slopes by condition, groups at their own mean ideology
fit_ind_C <- lm(individualizing ~ group * condition + ideology_c * condition, data = data)
fit_bin_C <- lm(binding ~ group * condition + ideology_c * condition, data = data)
emm_C <- function(fit) emmeans(fit, ~ group * condition,
                               cov.reduce = ideology_c ~ group)

rob <- rbind(
  h2_under(fit_ind, emm_ind, "Individualizing", "A. Preregistered"),
  h2_under(fit_ind_B, emmeans(fit_ind_B, ~ group * condition), "Individualizing", "B. No covariate"),
  h2_under(fit_ind_C, emm_C(fit_ind_C), "Individualizing", "C. Separate slopes"),
  h2_under(fit_bin, emm_bin, "Binding", "A. Preregistered"),
  h2_under(fit_bin_B, emmeans(fit_bin_B, ~ group * condition), "Binding", "B. No covariate"),
  h2_under(fit_bin_C, emm_C(fit_bin_C), "Binding", "C. Separate slopes")
)
rob <- rob[order(rob$family, rob$contrast, rob$spec), ]
kable(rob, digits = 3, row.names = FALSE,
      caption = "H2 accuracy contrasts across specifications (Delta = metaperception - actual)")
```

Table 12: H2 accuracy contrasts across specifications (Delta = metaperception - actual)

spec	family	contrast	Delta	SE	t	p
A. Preregistered	Binding	Conservative outgroup vs Liberal metaperception	-0.524	0.164	3.187	.002
B. No covariate	Binding	Conservative outgroup vs Liberal metaperception	-0.735	0.133	5.545	< .001
C. Separate slopes	Binding	Conservative outgroup vs Liberal metaperception	-0.729	0.131	5.567	< .001
A. Preregistered	Binding	Liberal outgroup vs Conservative metaperception	0.064	0.166	-0.387	.699
B. No covariate	Binding	Liberal outgroup vs Conservative metaperception	0.278	0.134	-2.073	.039
C. Separate slopes	Binding	Liberal outgroup vs Conservative metaperception	0.287	0.132	-2.165	.031
A. Preregistered	Individualizing	Conservative outgroup vs Liberal metaperception	-0.169	0.153	1.106	.269
B. No covariate	Individualizing	Conservative outgroup vs Liberal metaperception	-0.266	0.123	2.161	.031
C. Separate slopes	Individualizing	Conservative outgroup vs Liberal metaperception	-0.270	0.123	2.191	.029
A. Preregistered	Individualizing	Liberal outgroup vs Conservative metaperception	0.748	0.155	-4.833	< .001
B. No covariate	Individualizing	Liberal outgroup vs Conservative metaperception	0.846	0.125	-6.786	< .001
C. Separate slopes	Individualizing	Liberal outgroup vs Conservative metaperception	0.837	0.125	-6.714	< .001

9. The `mixed_manova` designs, corrected

```
two_by_two <- function(conds, label) {
  d <- droplevels(subset(data, condition %in% conds))
  d$ideology_c <- d$ideology - mean(d$ideology, na.rm = TRUE)
  rbind(
    cbind(design = label, DV = "Individualizing",
          ancova_table(lm(individualizing ~ group * condition + ideology_c, data = d), "")[, -1]),
    cbind(design = label, DV = "Binding",
          ancova_table(lm(binding ~ group * condition + ideology_c, data = d), "")[, -1])
  )
}
kable(rbind(two_by_two(c("Outgroup", "Metaperception"), "Outgroup vs Metaperception"),
            two_by_two(c("Outgroup", "Ingroup"), "Outgroup vs Ingroup")),
      digits = 3, row.names = FALSE)
```

design	DV	effect	F	df1	df2	p	eta2_p
Outgroup vs Metaperception	Individualizing	group	9.019	1	370	.003	0.024
Outgroup vs Metaperception	Individualizing	condition	9.826	1	370	.002	0.026
Outgroup vs Metaperception	Individualizing	ideology_c	0.726	1	370	.395	0.002
Outgroup vs Metaperception	Individualizing	group:condition	34.219	1	370	< .001	0.085
Outgroup vs Metaperception	Binding	group	10.903	1	370	.001	0.029
Outgroup vs Metaperception	Binding	condition	6.108	1	370	.014	0.016
Outgroup vs Metaperception	Binding	ideology_c	0.002	1	370	.961	0.000
Outgroup vs Metaperception	Binding	group:condition	129.847	1	370	< .001	0.260
Outgroup vs Ingroup	Individualizing	group	0.018	1	380	.893	0.000
Outgroup vs Ingroup	Individualizing	condition	58.732	1	380	< .001	0.134
Outgroup vs Ingroup	Individualizing	ideology_c	2.993	1	380	.084	0.008
Outgroup vs Ingroup	Individualizing	group:condition	121.319	1	380	< .001	0.242
Outgroup vs Ingroup	Binding	group	0.361	1	380	.548	0.001
Outgroup vs Ingroup	Binding	condition	0.011	1	380	.916	0.000
Outgroup vs Ingroup	Binding	ideology_c	2.369	1	380	.125	0.006
Outgroup vs Ingroup	Binding	group:condition	42.496	1	380	< .001	0.101

10. Reproduction checks against the manuscript

```
pillai_int <- summary(manova_seq, test = "Pillai")$stats["group:condition", "Pillai"]
h2 <- rbind(h2_ind$table, h2_bin$table)

checks <- data.frame(
  quantity = c("Pillai, group x condition",
               "Indiv: lib outgroup - con meta", "Indiv: con outgroup - lib meta",
               "Bind: lib outgroup - con meta", "Bind: con outgroup - lib meta",
               "d gap, individualizing", "d gap, binding"),
  manuscript = c(.371, -0.75, 0.17, -0.06, 0.52, -1.08, -0.65),
  computed = c(pillai_int, h2$M_diff, h2_ind$gap$d_gap, h2_bin$gap$d_gap)
)
checks$match <- abs(round(checks$computed, 2) - checks$manuscript) < .006
kable(checks, digits = 3)
```

quantity	manuscript	computed	match
Pillai, group x condition	0.371	0.371	TRUE
Indiv: lib outgroup - con meta	-0.750	-0.748	TRUE
Indiv: con outgroup - lib meta	0.170	0.169	TRUE
Bind: lib outgroup - con meta	-0.060	-0.064	TRUE
Bind: con outgroup - lib meta	0.520	0.524	TRUE
d gap, individualizing	-1.080	-1.083	TRUE
d gap, binding	-0.650	-0.647	TRUE

11. Figure 3

Ideology-adjusted scores (unit-preserving adjustment to the covariate mean, as in `morality`), with model-based EMMs ± 95% CI. `coord_cartesian` zooms without discarding observations, so violins use all the data.

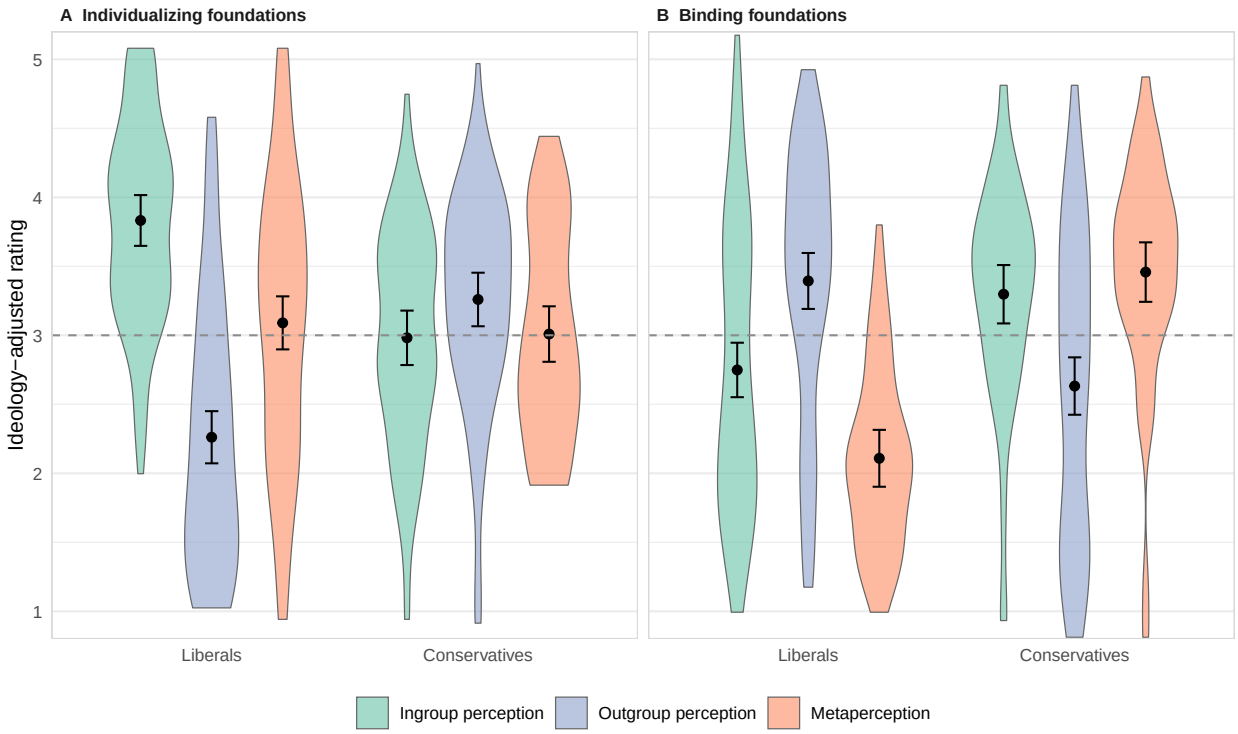
```
b_ind <- coef(fit_ind)[["ideology_c"]]
b_bin <- coef(fit_bin)[["ideology_c"]]

plot_df <- data %>%
  mutate(Individualizing = individualizing - b_ind * ideology_c,
         Binding = binding - b_bin * ideology_c) %>%
  select(group, condition, Individualizing, Binding) %>%
  pivot_longer(c(Individualizing, Binding), names_to = "family", values_to = "value")

emm_df <- rbind(
  cbind(family = "Individualizing", as.data.frame(emm_ind)),
  cbind(family = "Binding", as.data.frame(emm_bin))
)

cond_levels <- c("Ingroup", "Outgroup", "Metaperception")
cond_labels <- c("Ingroup perception", "Outgroup perception", "Metaperception")
plot_df$condition <- factor(plot_df$condition, cond_levels, cond_labels)
emm_df$condition  <- factor(emm_df$condition,  cond_levels, cond_labels)
fam_levels <- c("Individualizing", "Binding")
plot_df$family <- factor(plot_df$family, fam_levels,
                         c("A  Individualizing foundations", "B  Binding foundations"))
emm_df$family  <- factor(emm_df$family, fam_levels,
                         c("A  Individualizing foundations", "B  Binding foundations"))

pd <- position_dodge(width = 0.8)
ggplot(plot_df, aes(group, value, fill = condition)) +
  geom_violin(position = pd, width = 0.75, alpha = 0.6, colour = "grey40",
              linewidth = 0.3) +
  geom_errorbar(data = emm_df, aes(x = group, ymin = lower.CL, ymax = upper.CL,
                                   group = condition),
                inherit.aes = FALSE, position = pd, width = 0.15) +
  geom_point(data = emm_df, aes(x = group, y = emmean, group = condition),
             inherit.aes = FALSE, position = pd, size = 2) +
  geom_hline(yintercept = 3, linetype = "dashed", colour = "grey55") +
  facet_wrap(~ family) +
  scale_fill_manual(values = c("#66c2a5", "#8da0cb", "#fc8d62"), name = NULL) +
  coord_cartesian(ylim = c(1, 5)) +
  labs(x = NULL, y = "Ideology-adjusted rating") +
  theme_minimal(base_size = 11) +
  theme(legend.position = "bottom", strip.text = element_text(face = "bold", hjust = 0),
        panel.grid.major.x = element_blank(),
        panel.border = element_rect(fill = NA, colour = "grey85"))
```



12. Manuscript-ready text

```
pil <- summary(manova_seq, test = "Pillai")$stats["group:condition", ]
cat("**Interaction:** `Pillai's Trace $=", fmt_n(pil["Pillai"], 3), "$, $F(",
    pil["num Df"], ", ", pil["den Df"], ") = ", sprintf("%.2f", pil["approx F"]),
    "$, $p ", ifelse(pil["Pr(>F)"] < .001, "< .001", paste("=", fmt_n(pil["Pr(>F)"], 3))),
    "$`\n\n", sep = "")
```

Interaction: `Pillai's Trace $=.371$, $F(4, 1118) = 63.65$, $p < .001$`

```
cat("**Table 1 (Study 3) rows:**\n\n")
```

Table 1 (Study 3) rows:

```
for (i in seq_len(nrow(t1))) {
  r <- t1[i, ]
  tval <- if (r$reading == "Accurate") "---" else sprintf("%.2f (%d)", abs(r$t), r$df)
  cat(sprintf("- `%s & %s & $%s%.2f$ & %s & %s & %s \\\\`\n",
              r$judge, r$family, ifelse(r$Delta >= 0, "+", "-"), abs(r$Delta),
              tval, sub("< ", "$<$ ", r$p), r$reading))
}
```

- `Conservative & Individualizing & $+0.75$ & 4.83 (559) & $<$ .001 & Overestimate \\`
- `Liberal & Individualizing & $-0.17$ & --- & .269 & Accurate \\`
- `Conservative & Binding & $+0.06$ & --- & .699 & Accurate \\`
- `Liberal & Binding & $-0.52$ & 3.19 (559) & .002 & Underestimate \\`

Session info

```
sessionInfo()
```

```
R version 4.6.0 (2026-04-24)
Platform: aarch64-apple-darwin23
Running under: macOS Tahoe 26.5.2

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib;  LAPACK version 3.12.1

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Los_Angeles
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] knitr_1.51       effectsize_1.0.2 emmeans_2.0.3    car_3.1-5
[5] carData_3.0-6    ggplot2_4.0.3    tidyr_1.3.2      dplyr_1.2.1
[9] readxl_1.5.0.1

loaded via a namespace (and not attached):
 [1] generics_0.1.4     rgl_1.3.36         digest_0.6.39      magrittr_2.0.5
 [5] evaluate_1.0.5     grid_4.6.0         estimability_2.0.0 RColorBrewer_1.1-3
 [9] mvtnorm_1.4-1      fastmap_1.2.0      cellranger_1.1.0   jsonlite_2.0.0
[13] backports_1.5.1    Formula_1.2-5      purrr_1.2.2        scales_1.4.0
[17] abind_1.4-8        cli_3.6.6          rlang_1.2.0        base64enc_0.1-6
[21] withr_3.0.3        yaml_2.3.12        otel_0.2.0         tools_4.6.0
[25] datawizard_1.3.1   heplots_1.8.1      bayestestR_0.18.1  broom_1.0.13
[29] vctrs_0.7.3        R6_2.6.1           lifecycle_1.0.5    htmlwidgets_1.6.4
[33] MASS_7.3-65        insight_1.5.2      pkgconfig_2.0.3    pillar_1.11.1
[37] gtable_0.3.6       glue_1.8.1         xfun_0.59          tibble_3.3.1
[41] tidyselect_1.2.1   rstudioapi_0.19.0  parameters_0.29.2  farver_2.1.2
[45] htmltools_0.5.9    labeling_0.4.3     rmarkdown_2.31     compiler_4.6.0
[49] S7_0.2.2
```